Issac Roy

U.S. Citizen, San Diego, CA

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Education

University of California San Diego (3.97 GPA) **March 2027**

B.S. Data Science | *Math Minor* | *Cognitive Science Minor* *San Diego, CA*

Experience

Amazon Web Services | *Software Development Engineer Intern* | *EBS Snapshots* **September 2026 – Present**

- Built and deployed an automated pipeline on AWS (CDK, Fargate, EventBridge, DynamoDB, S3) that keeps on-call agents' knowledge bases in sync with their sources, such as code, team docs and PDF/DOCX files.
- Designed an extensible connector framework with a shared test suite, developed a change-significance classifier that blocks cosmetic changes, and gated every knowledge update behind regression tests.

Apple | *Software Engineer Intern* | *WebKit* **June 2026 – September 2026**

- Implemented `flex-wrap:balance`, a new CSS feature, in WebKit using dynamic programming to minimize line-length imbalance, and built an interactive visualizer to compare layouts.
- Developed tooling that uses WebKit's test history to separate flaky failures from real regressions, clearing false positives from pre-commit testing and unblocking 50+ PRs per week.

Neurocrine Biosciences | *Data Engineer Intern* | *Clinical Solutions* **June 2025 – September 2025**

- Saved $100k annually by architecting a serverless containerized data ingestion pipeline on AWS (ECS, Lambda, S3, Redshift), eliminating dependency on third-party vendors.
- Saved 50+ hours / month for study teams by developing a data pipeline with LLM endpoints from AWS Bedrock to convert raw patient data into study narratives.
- Implemented DevOps workflows using Terraform (IaC), Github Actions, and multi-environment CI/CD pipelines.

Projects

Turbo 3D Engine | *C++, SDL2, Linear Algebra* **March 2025**

- Built a 3D graphics engine from scratch, implementing a custom rasterization pipeline, matrix transformations for camera space, custom material system, texture mapping, and multiple lighting models without external graphics libraries.
- Wrote an OBJ file parser to load and render 3D models from file and optimized performance through efficient vertex processing algorithms and a custom linear algebra library.

GhostWriter | *TypeScript, Bun, React, SQLite, LaTeX* **October 2026 – Present**

- Building a resume generator that scrapes a LinkedIn profile through a browser extension into SQLite with change history, then compiles a LaTeX resume tailored to each job posting.

Neural Nector | *React, TypeScript, PyTorch, Python, FastAPI, Postgres* **December 2025**

- Developed a full-stack web app game that had 100+ players, featuring a React frontend and a FastAPI backend.
- Optimized a custom GAN model, managing async tasks and database transactions to track game states & leaderboards.

Project Daedalus (G-Class Rocket) | *C++, KiCad, Onshape, Arduino* **October 2025 – Present**

- Developed a custom PCB, integrating an MPU6050, BMP388, and LoRa radio for telemetry and black-box logging.
- Designed & 3D-printed avionics bay to sustain high-G loads and protect electronics, targeting a 5,500ft (1 mi+) apogee.

Leadership

AWS Student Builder Group at UCSD | *Founder & Technical Lead* **November 2025 – Present**

- Founded AWS SBG to teach students cloud computing, serverless architecture, and industry best practices.
- Leading technical workshops on core AWS services, equipping 200+ members with practical cloud skills.

Data Science Student Society (DS3) | *Software Lead* **October 2024 – June 2026**

- Led a 10-person team to build a membership portal for 1,000+ students. Established automated CI/CD pipelines via GitHub Actions and introduced Agile methodologies to improve productivity and accountability.
- Developed the website frontend and engineered a Supabase backend, incorporating auth and schemas for member tracking, event attendance, and analytics.

Technical Skills

Languages: C++, C, Python, Java, SQL, TypeScript, JavaScript, Bash, HTML/CSS
Tools: AWS, React, Terraform, Docker, Git, CI/CD, Postgres, MongoDB, Raspberry Pi, Linux
Certifications: AWS Cloud Practitioner, AWS AI Practitioner, AWS Solutions Architect Associate